Course notes

Definitions

$u_n \to \ell \in \mathbb{R}$ means that any open interval containing ℓ contains all terms from a
It is said that (u_n) converges to ℓ. $u_n \to +\infty$ means that for all real A, there is
one row N such that $n \geq N \Rightarrow u_n \geq A$. $u_n \to -\infty$ means that for all real A, there is N such that
$n \geq N \Rightarrow u_n \leq A$.

A sequence may have no limit, even infinity: $(-1)^n$ oscillation between -1 and 1. Any further action
converging is bounded, but the converse is false.

Operations and comparisons

For two converging sequences of limits a, b, the sum tends to $a+b$, the product ab and quotient $\frac{a}{b}$ if $b \neq 0$. Rules with infinity are used by controlling signs.
$+\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$ do not allow for direct conclusion.

Example. $\frac{3n^2-n+1}{2n^2+4} = \frac{3-\frac{1}{n}+\frac{1}{n^2}}{2+\frac{4}{n^2}} \to \frac{3}{2}$. For $\sqrt{n^2+n}-n$, multiply by the conjugate quantity given $\frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2}$.

Theorems

If $u_n \leq v_n$ from a certain rank and $u_n \to +\infty$, then $v_n \to +\infty$. If $a_n \leq u_n \leq b_n$ from a certain rank and $a_n, b_n \to \ell$, then $u_n \to \ell$ (gendarmes) Any increasing and increased succession converges. Any decreasing and minor sequence converges. This theorem is allowed.

Demonstration by comparison. For every A, the terms u exceeding A beyond a row N_1. Inequality $v_n \geq u_n$ equals beyond a row N_2. For $n \geq \max(N_1, N_2)$, $v_n \geq A$.

Demonstration: increasing not increased. For every A, the non-majority provides N such that $u_N > A$. By growth, $n \geq N \Rightarrow u_n \geq u_N > A$, where $u_n \to +\infty$. Non-majority alone is not enough.

Geometrical sequences

$$q^n \longrightarrow \begin{cases} 0 & \text{if } -1 < q < 1, \\ 1 & \text{if } q = 1, \\ +\infty & \text{if } q > 1. \end{cases}$$

If $q \leq -1$, (q^n) no limit: alternation for $q = -1$; paired and odd sub-sequences of signs and unbounded modules for $q < -1$.

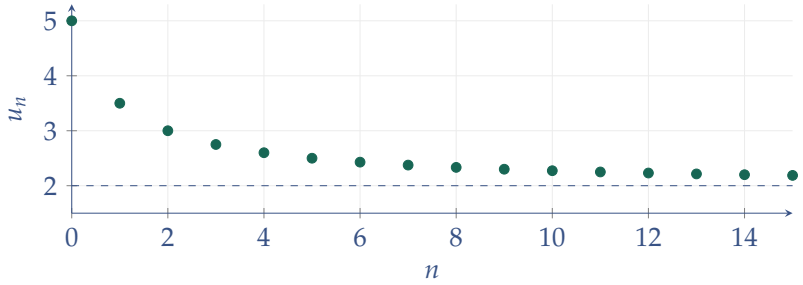
Bernoulli and geometric limit. Fixes $a \geq 0$. Demontrons $P(n) : \text{« } (1+a)^n \geq 1+na$ In row 0, the two members are worth 1. $n \geq 0$ and suppose $P(n)$ Like one. $+a > 0$,

$$(1+a)^{n+1} \geq (1+na)(1+a) = 1 + (n+1)a + na^2 \geq 1 + (n+1)a.$$

It's $P(n+1)$ since $na^2 \geq 0$. By induction, inequality is true for all $n \geq 0$. If $q > 1$, take $a = q - 1 > 0$ and compare to $1 + na$. If $0 < q < 1$, apply this result to $\frac{1}{q}$Then take the reverses. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ and the gendarmes conclude. $q = 0$ deals from row 1.

Recursively defined sequence

For $u_{n+1} = f(u_n)$, establish a bound and monotonicity to prove convergence. f is continues in the limit ℓ, the passage to the limit gives $\ell = f(\ell)$ (continuity discussed in Chapter 4). Solving this equation never proves, on its own, that the sequence converges.



Points close to 2 without reaching: $u_n = 2 + \frac{3}{n+1}$.

Give meaning to the limits

2.1 One band and all subsequent ranks

First use: S14.

Definition · Converging to ℓ

Next (u_n) converges to ℓ if, for any accuracy $\varepsilon > 0$, there is a row N such that $n \geq N$ implies $|u_n - \ell| < \varepsilon$. The rank depends on the accuracy, but once chosen, it is suitable for all the following terms.

Solved example · Calculate a valid rank

For $u_n = 2 + \frac{3}{n+1}$, $|u_n - 2| = \frac{3}{n+1}$. Precision 0 ,1 requires $n + 1 > 30$, so $n \geq 30$. Rank 29 is not suitable: the difference is exactly 0 ,1. For any precision, any N such that $N + 1 > 3/\varepsilon$ It's okay.

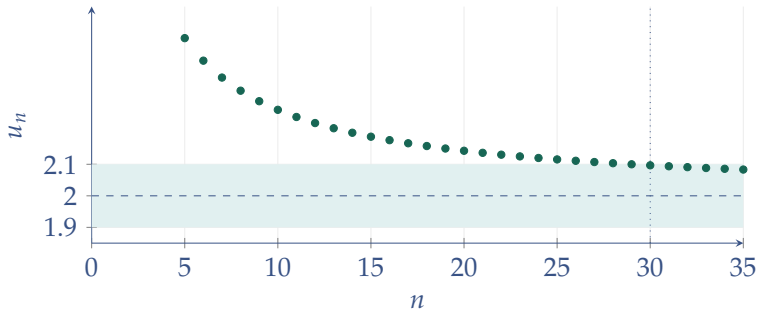


Figure 2.1. Band $]1{,}9;2{,}1[$: all terms of rank $n \geq 30$ stay there.

A1 · Immediate application For $v_n = 5 - \frac{2}{n+1}$, determine the first ranking from which $4{,}95 < v_n < 5.05$. Check the previous row and explain why all of the following agree.

A2 · Counterexample "There is a term in each band around 0, so the sequence converges to 0. » Refuse with a sequence whose even terms are null and the odd terms are worth 1.

2.2 Transform before concluding

First use: S16 and S19.

Solved Example · A Form Does Not Set a Limit

The three quotients n/n, n/n^2 and n^2/n all have a shape ∞/∞. Their limits are however 1, 0 and $+\infty$. For $\frac{3n^2-n+1}{2n^2+4}$ divide numerator and denominator by n^2 gives the limit $\frac{3}{2}$. The transformation is valid for $n \geq 1$, which is enough for a limit.

A3 · Choose before calculating Determining the limits of $\frac{2n+3}{n^2+1}$, $\sqrt{n^2+n}-n$ and $\frac{(-1)^n}{n+1}$. Name, for Each, transformation or decisive theorem.

A first step: In the second expression, multiply by the conjugate quantity. In the third, frame the absolute value.

2.3 Converging and then identifying

First use: S20; continuity identification after S32.

Ownership · Monotonous Convergence · Accepted

A increasing and increased sequence converges towards a real. The theorem does not say that it reaches its or that its limit is any of its majors.

Example Solved · A complete string

Let $u_0 = 1$ and $u_{n+1} = \sqrt{2+u_n}$. The interval $[1;2]$ is stable because $1 \leq u_n \leq 2$ results $\sqrt{3} \leq u_{n+1} \leq 2$. Over this interval, $2+x-x^2 = (2-x)(x+1) \geq 0$, therefore $u_{n+1} \geq u_n$ After comparison positive squares. The sequence converges to $\ell \in [1;2]$. After continuity, $\ell = \sqrt{2+\ell}$ causes $(\ell-2)(\ell+1) = 0$; only $\ell = 2$ belongs to $[1;2]$.

A4 · Repairing a Common difference / ratio Next $a_0 = 1$, $a_{n+1} = 2a_n$ checks a relationship whose only point 0. A student concludes $a_n \to 0$. Show his error and determine the actual behavior.

A5 · Independent evidence Let $b_0 = 0$, $b_{n+1} = \frac{1}{3}b_n + 2$. Prove $0 \leq b_n < 3$, then growth. deduce convergence and determine the limit by the explicit formula or, after S32, by continuity.

2.4 Don't forget the negative reasons

First use: S19 and S21.

Property · Behaviour of q^n

If $|q| < 1$, $q^n \to 0$. If $q > 1$, $q^n \to +\infty$. For $q = 1$, the sequence is constant. $q \leq -1$, it does not no limit, even infinite: the signs alternate, and the two sequences of even and odd ranks have incompatible behaviors.

Demonstration. For $q > 1$, write $q = 1+a$ with $a > 0$. Recurrence gives $(1+a)^n \geq 1+na$, which tends to $+\infty$; the comparison concludes. $< q < 1$, apply this result to $1/q$ and take The other way around. $-1 < q < 0$, $|q^n| = |q|^n \to 0$ results $q^n \to 0$ per frame. $q = 0$ is immediate from row 1. □

Example Solved · Common difference / ratio and Behaviour

For $q = -2$, the same terms 4^k tend to $+\infty$ and oddness -2×4^k To $-\infty$: None common limit. $q = -1/2$, the signs also alternate, but the module 2^{-n} tend towards 0; the sequence converges.

A6 · Compare properties For $(-1)^n$, $-\frac{1}{n+1}$ and n, indicate: bounded ? monotonous ? convergent ? Give justifications, not just a table of answers.

Wait a minute at the courthouse. Explain why "increasing" does not mean "positive" or "trend towards $+\infty$ ».

Review postponed · S33 then S66. Re-identification by continuity; for a complete sequence, reconstruct the same bound chain, monotonicity, convergence, identification.

Research activity

A limit is not a last term · 30 min.

What's next? (u_n) defined on $\mathbb{N}$ by $u_n = 2 + \frac{3}{n+1}$.

1. Calculate u_0, u_9 and u_{99}. Is there an integer n such that $u_n = 2$? Justify.

2. Determine the smallest integer N such as, for the integer $n \geq N$, $|u_n - 2| < 0{,}01$.

3. Either $\varepsilon > 0$. Give, depending on ε, one integer N such that $|u_n - 2| < \varepsilon$ for every $n \geq N$.

4. We define on $\mathbb{N}$ the rest (v_n) by $v_n = 2 + \frac{(-1)^n}{n+1}$. Demonstrate that it converges to 2 and that What does this example show?